

1. Problem Statement

Aerofit, a fitness equipment brand, sells three treadmill models: KP281 (entry), KP481 (mid), KP781 (premium). The market research team wants to:

- Profile customers for each treadmill.
- Investigate whether customer characteristics (age, gender, income, fitness, marital status, usage, miles) influence treadmill choice.
- Compute descriptive statistics & probabilities to help in better product recommendation.

```
import pandas as pd

file_id = "1mAfjlG0kXCawx-XIzLEy0jlmq_c1Gul3" # from your link
url = f"https://drive.google.com/uc?export=download&id={file_id}"

df = pd.read_csv(url, sep=",")
df
```

	Product	Age	Gender	Education	MaritalStatus	Usage	Fitness	Income	Miles
0	KP281	18	Male	14	Single	3	4	29562	112
1	KP281	19	Male	15	Single	2	3	31836	75
2	KP281	19	Female	14	Partnered	4	3	30699	66
3	KP281	19	Male	12	Single	3	3	32973	85
4	KP281	20	Male	13	Partnered	4	2	35247	47
...	...	...	...	...	...	...	...	...	...
175	KP781	40	Male	21	Single	6	5	83416	200
176	KP781	42	Male	18	Single	5	4	89641	200
177	KP781	45	Male	16	Single	5	5	90886	160
178	KP781	47	Male	18	Partnered	4	5	104581	120
179	KP781	48	Male	18	Partnered	4	5	95508	180

180 rows x 9 columns

Next steps: [Generate code with df](#) [New interactive sheet](#)

2. Dataset & Basic Checks

```
# 2.1.Shape of dataset
print("Shape of dataset:", df.shape)
```

Shape of dataset: (180, 9)

```
# 2.2.Column data types and non-null counts
print("\nDataset Info:")
print(df.info())
```

```
Dataset Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 180 entries, 0 to 179
Data columns (total 9 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Product         180 non-null   object
1   Age             180 non-null   int64
2   Gender          180 non-null   object
3   Education       180 non-null   int64
4   MaritalStatus   180 non-null   object
5   Usage           180 non-null   int64
6   Fitness         180 non-null   int64
7   Income          180 non-null   int64
8   Miles           180 non-null   int64
dtypes: int64(6), object(3)
memory usage: 12.8+ KB
None
```

```
# 2.3.Summary statistics for numerical columns
print("\nSummary statistics (numerical):")
display(df.describe().T)
```



Summary statistics (numerical):

	count	mean	std	min	25%	50%	75%	max	
Age	180.0	28.788889	6.943498	18.0	24.00	26.0	33.00	50.0	
Education	180.0	15.572222	1.617055	12.0	14.00	16.0	16.00	21.0	
Usage	180.0	3.455556	1.084797	2.0	3.00	3.0	4.00	7.0	
Fitness	180.0	3.311111	0.958869	1.0	3.00	3.0	4.00	5.0	
Income	180.0	53719.577778	16506.684226	29562.0	44058.75	50596.5	58668.00	104581.0	
Miles	180.0	103.194444	51.863605	21.0	66.00	94.0	114.75	360.0	

```
# 2.4.Summary statistics for categorical columns
print("\nSummary statistics (categorical):")
display(df.describe(include=['object']).T)
```

Summary statistics (categorical):

	count	unique	top	freq	
Product	180	3	KP281	80	
Gender	180	2	Male	104	
MaritalStatus	180	2	Partnered	107	

```
# 2.5.Value counts for categorical variables
print("\nProduct distribution:")
print(df['Product'].value_counts())

print("\nGender distribution:")
print(df['Gender'].value_counts())

print("\nMarital Status distribution:")
print(df['MaritalStatus'].value_counts())
```

Product distribution:
 Product
 KP281 80
 KP481 60
 KP781 40
 Name: count, dtype: int64

Gender distribution:
 Gender
 Male 104
 Female 76
 Name: count, dtype: int64

Marital Status distribution:
 MaritalStatus
 Partnered 107
 Single 73
 Name: count, dtype: int64

3. Missing Values & Outlier Detection

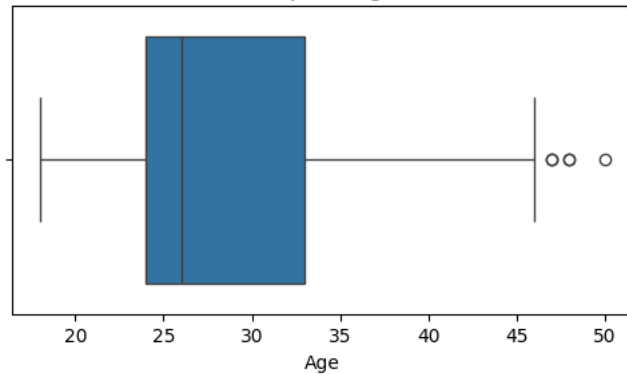
```
# 3.1.Check for missing values
print("Missing values per column:")
print(df.isnull().sum())
```

Missing values per column:
 Product 0
 Age 0
 Gender 0
 Education 0
 MaritalStatus 0
 Usage 0
 Fitness 0
 Income 0
 Miles 0
 dtype: int64

```
# 3.2.Boxplots for Outlier Detection
import seaborn as sns
import matplotlib.pyplot as plt
num_cols = ['Age', 'Education', 'Usage', 'Fitness', 'Income', 'Miles']
for col in num_cols:
```

```
plt.figure(figsize=(6,3))
sns.boxplot(x=df[col])
plt.title(f'Boxplot - {col}')
plt.show()
```


Boxplot - Age



Boxplot - Education

```
# 3.3.Compare mean, median, and skewness
for col in num_cols:
    mean_val = df[col].mean()
    median_val = df[col].median()
    skew_val = df[col].skew()
    print(f'{col}: mean={mean_val:.2f}, median={median_val:.2f}, skew={skew_val:.2f}')
```

```
Age: mean=28.79, median=26.00, skew=0.98
Education: mean=15.57, median=16.00, skew=0.62
Usage: mean=3.46, median=3.00, skew=0.74
Fitness: mean=3.31, median=3.00, skew=0.45
Income: mean=53719.58, median=50596.50, skew=1.29
Miles: mean=103.19, median=94.00, skew=1.72
```

```
# 3.4.Detect outliers using IQR method
def detect_outliers_iqr(series):
    Q1 = series.quantile(0.25)
    Q3 = series.quantile(0.75)
    IQR = Q3 - Q1
    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR
    return ((series < lower) | (series > upper)).sum(), lower, upper

for col in num_cols:
    outliers, low, up = detect_outliers_iqr(df[col])
    print(f'{col}: {outliers} outliers, bounds=({low:.2f}, {up:.2f})')
```

```
Age: 5 outliers, bounds=(10.50, 46.50)
Education: 4 outliers, bounds=(11.00, 19.00)
Usage: 9 outliers, bounds=(1.50, 5.50)
Fitness: 2 outliers, bounds=(1.50, 5.50)
Income: 19 outliers, bounds=(22144.88, 80581.88)
Miles: 13 outliers, bounds=(-7.12, 187.88)
```

Boxplot - Fitness

Missing values: None found.

Outliers detected:

Income: A few customers with very high incomes (> \$90,000).

Miles: Several heavy users (> 200 miles per week).

Mean vs Median: Small gaps for most features → distributions are fairly balanced, except Income and Miles which show positive skew (long right tail).

These outliers represent high-income premium buyers and serious fitness enthusiasts, so they may be genuine business cases rather than data errors.

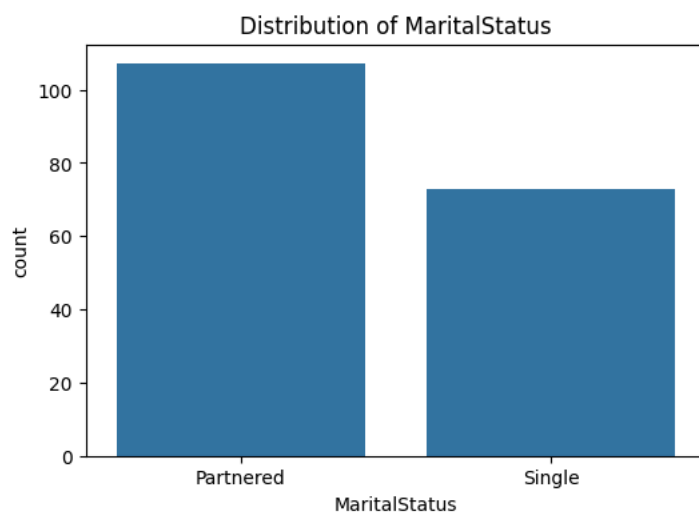
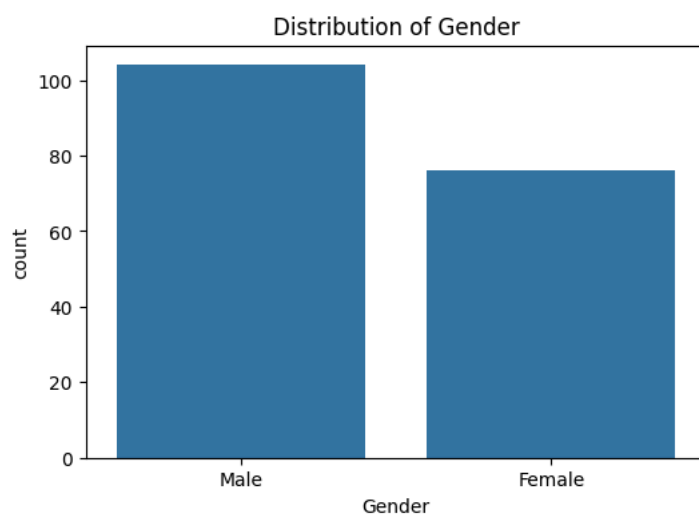
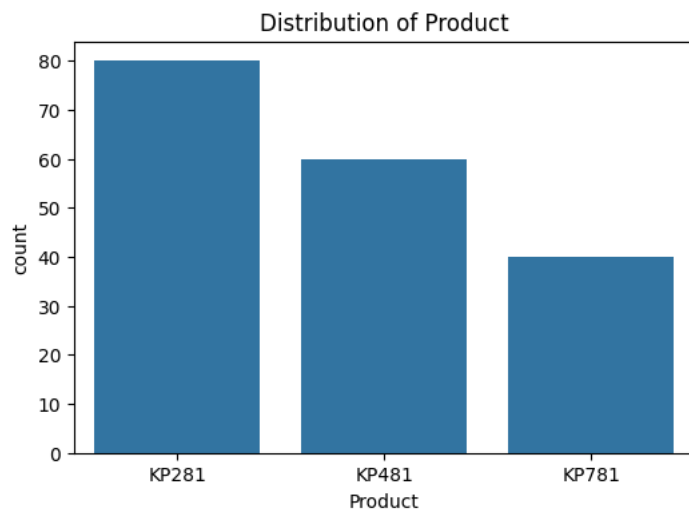
4. Univariate Analysis

Boxplot - Income

```
# 4.1.Categorical univariate analysis
categorical_cols = ['Product', 'Gender', 'MaritalStatus']

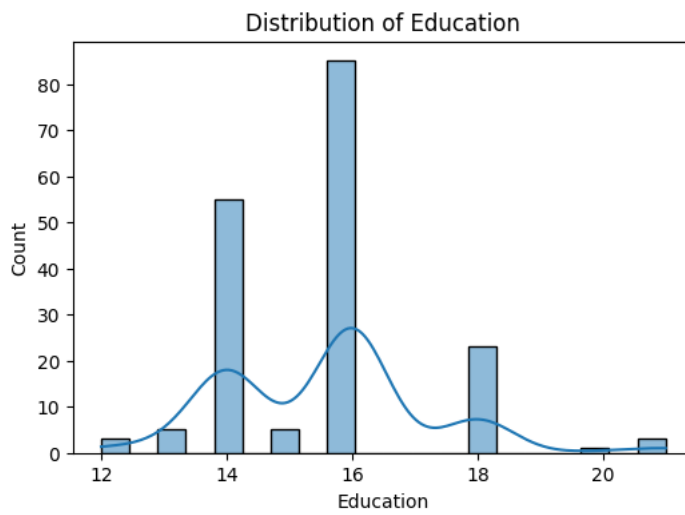
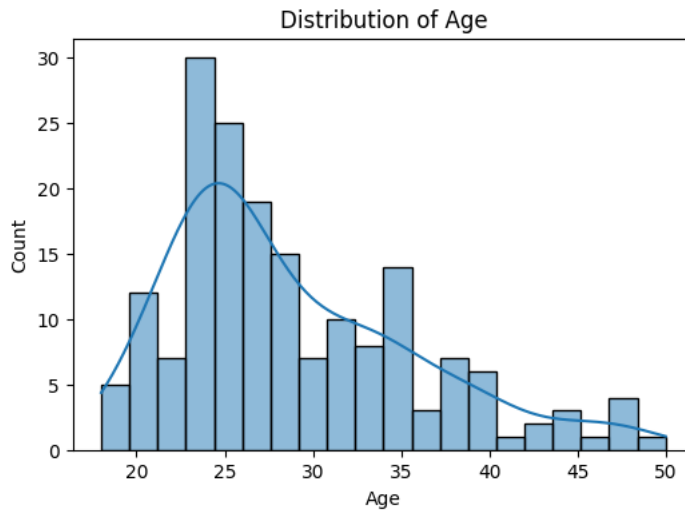
import seaborn as sns
import matplotlib.pyplot as plt

for col in categorical_cols:
    plt.figure(figsize=(6,4))
    sns.countplot(x=col, data=df, order=df[col].value_counts().index)
    plt.title(f'Distribution of {col}')
    plt.show()
```



```
# 4.2.Numerical univariate analysis
num_cols = ['Age','Education','Usage','Fitness','Income','Miles']

for col in num_cols:
    plt.figure(figsize=(6,4))
    sns.histplot(df[col], kde=True, bins=20)
    plt.title(f"Distribution of {col}")
    plt.show()
```

Distribution of Usage

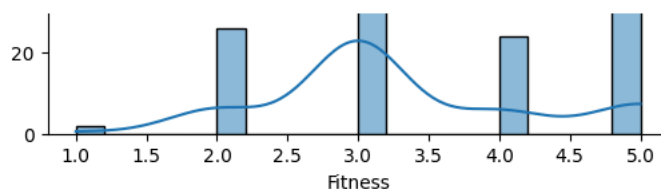
Observations – Univariate

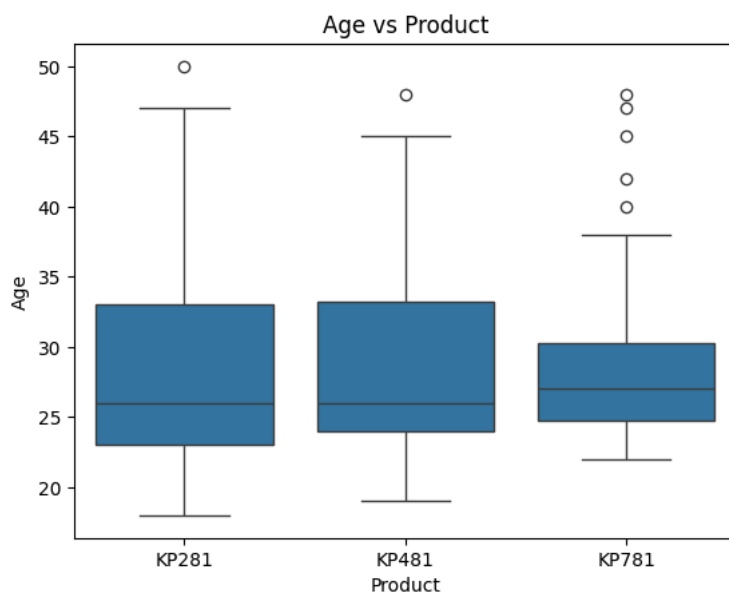
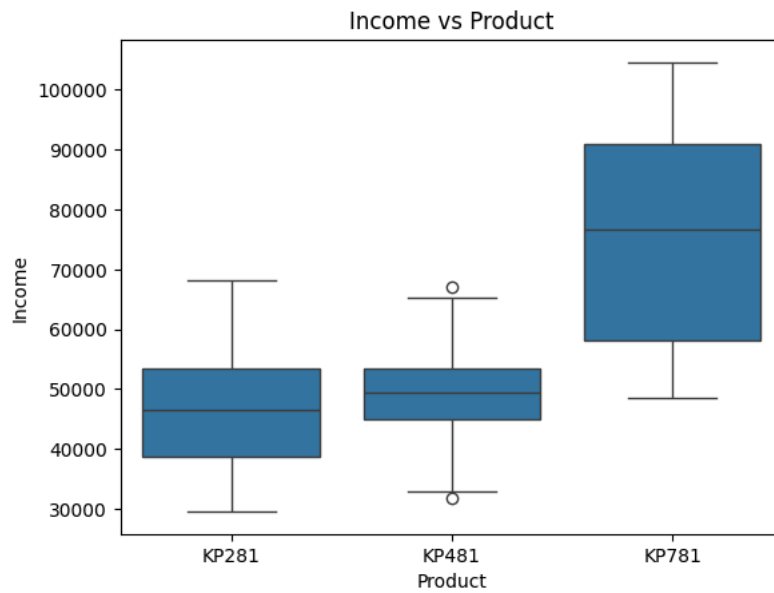
- Product:** KP281 (80 customers) is the most purchased treadmill, followed by KP481 (60) and KP781 (40). Entry-level and mid-range dominate overall sales.
- Gender:** Male customers (58%) are higher than female customers (42%), showing a male-dominated customer base.
- Marital Status:** Partnered customers (59%) are slightly higher than single customers (41%), indicating good penetration across both groups.
- Age:** Most customers are between 20–35 years, with a peak in the 24–28 age group. Very few buyers are above 40.
- Education:** Majority have 14–16 years of education, reflecting a well-educated customer base.
- Usage:** Most customers plan to use the treadmill 3–4 times per week, showing moderate usage intent.
- Fitness:** Fitness levels are centered around 3 (average), with fewer customers at extreme ratings (1 or 5).
- Income:** Average income is around \$50k, with a right-skewed distribution.
- Miles:** Average is ~100 miles per week, with most users in the 50–150 range. A few outliers run more than 250 miles, likely KP781 users.

4.2 Bivariate Analysis

```
# Boxplots
sns.boxplot(x='Product', y='Income', data=df)
plt.title("Income vs Product")
plt.show()

sns.boxplot(x='Product', y='Age', data=df)
plt.title("Age vs Product")
plt.show()
```





```
# Countplots
sns.countplot(x='Product', hue='Gender', data=df)
plt.title("Gender vs Product")
plt.show()

sns.countplot(x='Product', hue='MaritalStatus', data=df)
plt.title("Marital Status vs Product")
plt.show()
```